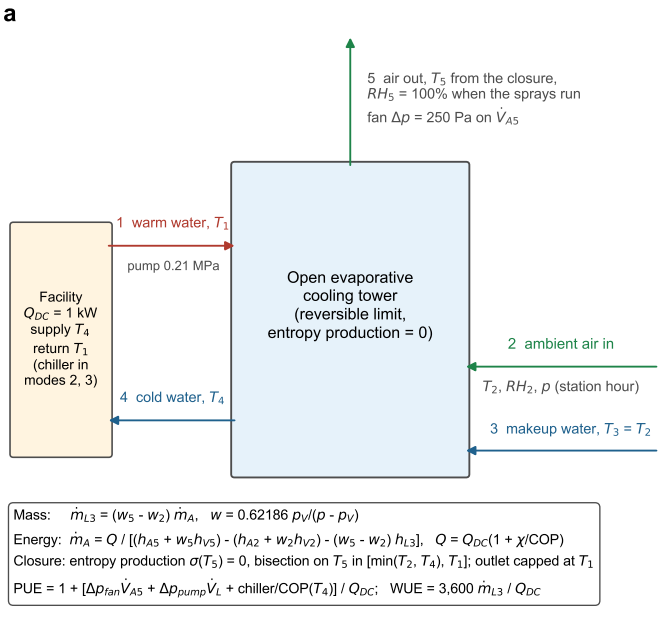
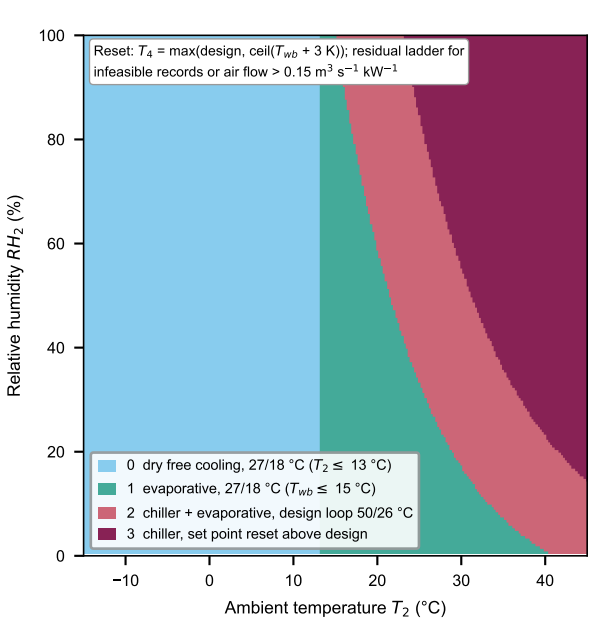
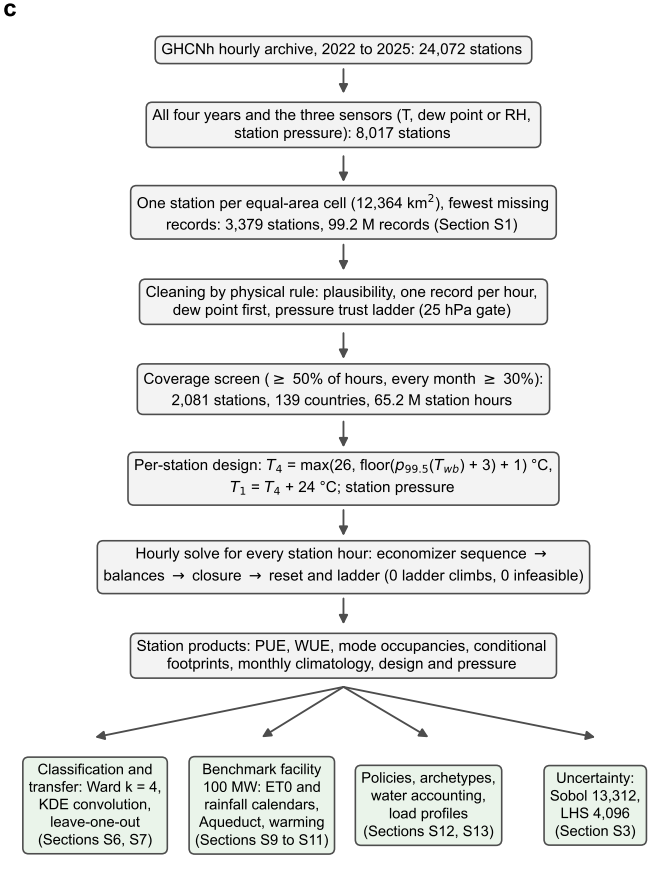
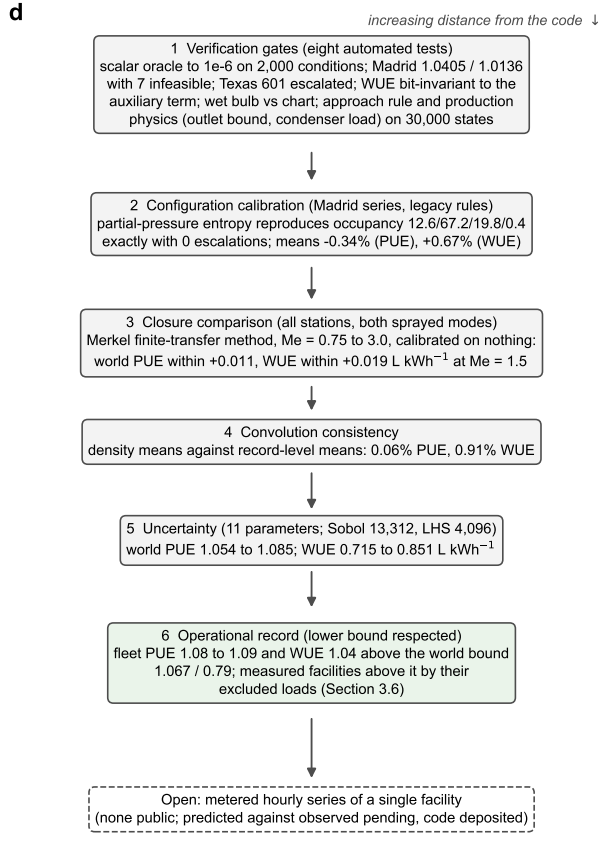

a
5 air out, T_5 from the closure,
RH_5 = 100% when the sprays run
fan Δp = 250 Pa on $\dot{V}_{A5}$
1 warm water, T_1
pump 0.21 MPa
Facility
Q_{DC} = 1 kW
supply T_4
return T_1
(chiller in
modes 2, 3)
Open evaporative
cooling tower
(reversible limit,
entropy production = 0)
2 ambient air in
T_2, RH_2, p (station hour)
4 cold water, T_4
3 makeup water, $T_3 = T_2$
Mass: $\dot{m}_{L3} = (w_5 - w_2)\,\dot{m}_A$, $w = 0.62186\,p_V/(p - p_V)$
Energy: $\dot{m}_A = Q / [(h_{A5} + w_5 h_{V5}) - (h_{A2} + w_2 h_{V2}) - (w_5 - w_2)\,h_{L3}]$, $Q = Q_{DC}(1 + \chi/\mathrm{COP})$
Closure: entropy production $\sigma(T_5) = 0$, bisection on T_5 in $[\min(T_2, T_4), T_1]$; outlet capped at T_1
PUE = $1 + [\Delta p_{fan}\dot{V}_{A5} + \Delta p_{pump}\dot{V}_L + \mathrm{chiller/COP}(T_4)] / Q_{DC}$; WUE = $3{,}600\,\dot{m}_{L3} / Q_{DC}$


b
Reset: T_4 = max(design, ceil(T_{wb} + 3 K)); residual ladder for
infeasible records or air flow > 0.15 m^3 s^{-1} kW^{-1}
0 dry free cooling, 27/18 °C ($T_2 \le$ 13 °C)
1 evaporative, 27/18 °C ($T_{wb} \le$ 15 °C)
2 chiller + evaporative, design loop 50/26 °C
3 chiller, set point reset above design
Relative humidity RH_2 (%)
Ambient temperature T_2 (°C)
0
20
40
60
80
100
−10
0
10
20
30
40


c
GHCNh hourly archive, 2022 to 2025: 24,072 stations
All four years and the three sensors (T, dew point or RH, station pressure): 8,017 stations
One station per equal-area cell (12,364 km^2), fewest missing records: 3,379 stations, 99.2 M records (Section S1)
Cleaning by physical rule: plausibility, one record per hour, dew point first, pressure trust ladder (25 hPa gate)
Coverage screen (≥ 50% of hours, every month ≥ 30%): 2,081 stations, 139 countries, 65.2 M station hours
Per-station design: T_4 = max(26, floor($p_{99.5}(T_{wb})$ + 3) + 1) °C, $T_1 = T_4$ + 24 °C; station pressure
Hourly solve for every station hour: economizer sequence → balances → closure → reset and ladder (0 ladder climbs, 0 infeasible)
Station products: PUE, WUE, mode occupancies, conditional footprints, monthly climatology, design and pressure
Classification and transfer: Ward k = 4, KDE convolution, leave-one-out (Sections S6, S7)
Benchmark facility 100 MW: ET0 and rainfall calendars, Aqueduct, warming (Sections S9 to S11)
Policies, archetypes, water accounting, load profiles (Sections S12, S13)
Uncertainty: Sobol 13,312, LHS 4,096 (Section S3)


d
increasing distance from the code ↓
1 Verification gates (eight automated tests)
scalar oracle to 1e-6 on 2,000 conditions; Madrid 1.0405 / 1.0136 with 7 infeasible; Texas 601 escalated; WUE bit-invariant to the auxiliary term; wet bulb vs chart; approach rule and production physics (outlet bound, condenser load) on 30,000 states
2 Configuration calibration (Madrid series, legacy rules)
partial-pressure entropy reproduces occupancy 12.6/67.2/19.8/0.4 exactly with 0 escalations; means -0.34% (PUE), +0.67% (WUE)
3 Closure comparison (all stations, both sprayed modes)
Merkel finite-transfer method, Me = 0.75 to 3.0, calibrated on nothing: world PUE within +0.011, WUE within +0.019 L kWh^{-1} at Me = 1.5
4 Convolution consistency
density means against record-level means: 0.06% PUE, 0.91% WUE
5 Uncertainty (11 parameters; Sobol 13,312, LHS 4,096)
world PUE 1.054 to 1.085; WUE 0.715 to 0.851 L kWh^{-1}
6 Operational record (lower bound respected)
fleet PUE 1.08 to 1.09 and WUE 1.04 above the world bound 1.067 / 0.79; measured facilities above it by their excluded loads (Section 3.6)
Open: metered hourly series of a single facility
(none public; predicted against observed pending, code deposited)